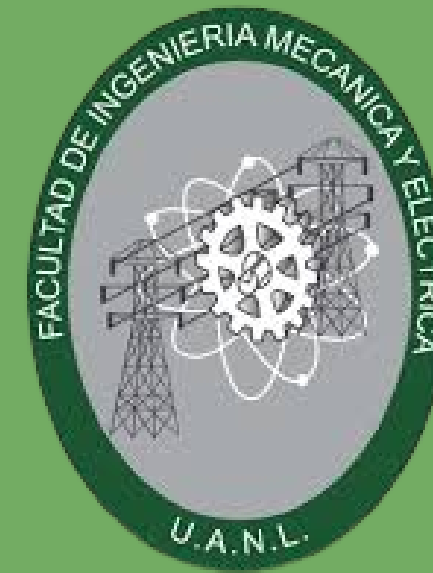




RCUANL

RoboCup Junior Rescue Maze



About us

RC-UANL is a team from Mexico, representing FIME and the Universidad Autónoma de Nuevo León, that competes in the Rescue Maze category at TMR 2026. It integrates robotics expertise with advanced technologies in autonomous systems to develop a robot capable of navigating complex environments, detecting victims, and overcoming obstacles, aiming to improve its performance throughout the navigation.

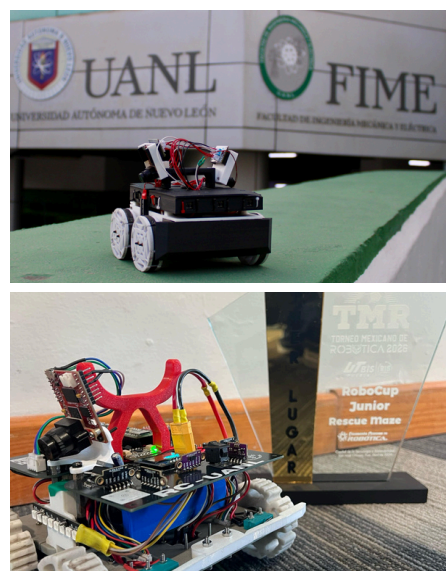
 **Diego Emilio Evangelista Gomez**
Designer
Responsible of mechanical design.

 **Stefany Giselle Colunga Rangel**
Programmer
Responsible of Mapping Algorithm

 **Rogelio Andrade López**
Electronics
Responsible of PCB design.

 **Axel Uriel Labrada Hernández**
Programmer
Responsible of victim detection.

Previous Results



- The team participated in the Robocup Mexican Open 2025, reaching a 3rd place at a national level.
- Winners of the Mexican Open 2026
- We are currently preparing to represent Mexico at RoboCup 2026, with the goal of reaching an international podium finish.

Platforms and Languages



More information



RC UANL

GITHUB

SOFTWARE

Mapping Algorithm

The robot's mapping system is based on a DFS (Depth-First Search) algorithm combined with an exploration strategy guided by absolute references, allowing for more consistent and predictable navigation within the maze. The environment is represented as a dynamic matrix, where each cell corresponds to a tile. As the robot advances, it records relevant information such as walls, available paths, detected victims, and special zones. This representation allows for the identification of explored and pending areas, optimizing decision-making. Unlike traditional strategies such as the "right-hand rule," the robot prioritizes its movements based on an absolute orientation system. An initial direction is defined as absolute north, determined by the robot's orientation at the start of the execution. From this reference, the exploration priority order is:

- Absolute North
- Absolute East
- Absolute South
- Absolute West

At each intersection, the algorithm evaluates the available directions following this order, always selecting the unexplored option with the highest priority. This allows for the generation of more uniform trajectories and facilitates the consistency of the generated map.

When the robot reaches a point with no new available routes, the algorithm executes a backtracking process, returning to previously visited nodes until it finds new unexplored branches, this ensuring complete coverage of the environment.

Track Strategy

In the first run, the robot avoids the red zone due to its high complexity, prioritizing stable navigation and returning to the starting point to obtain the exit bonus.

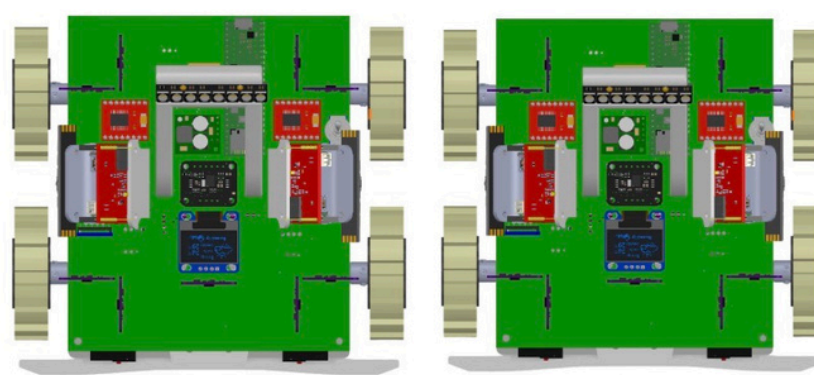
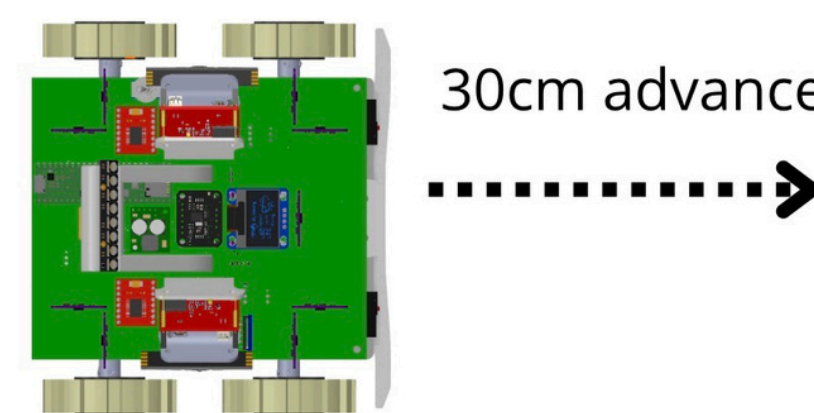
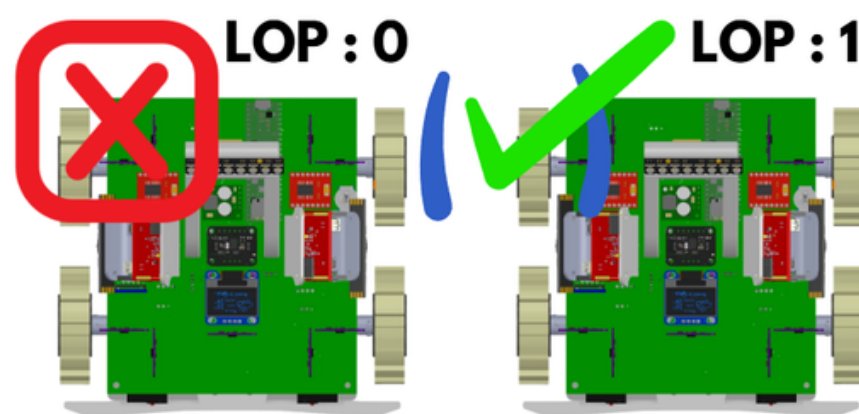
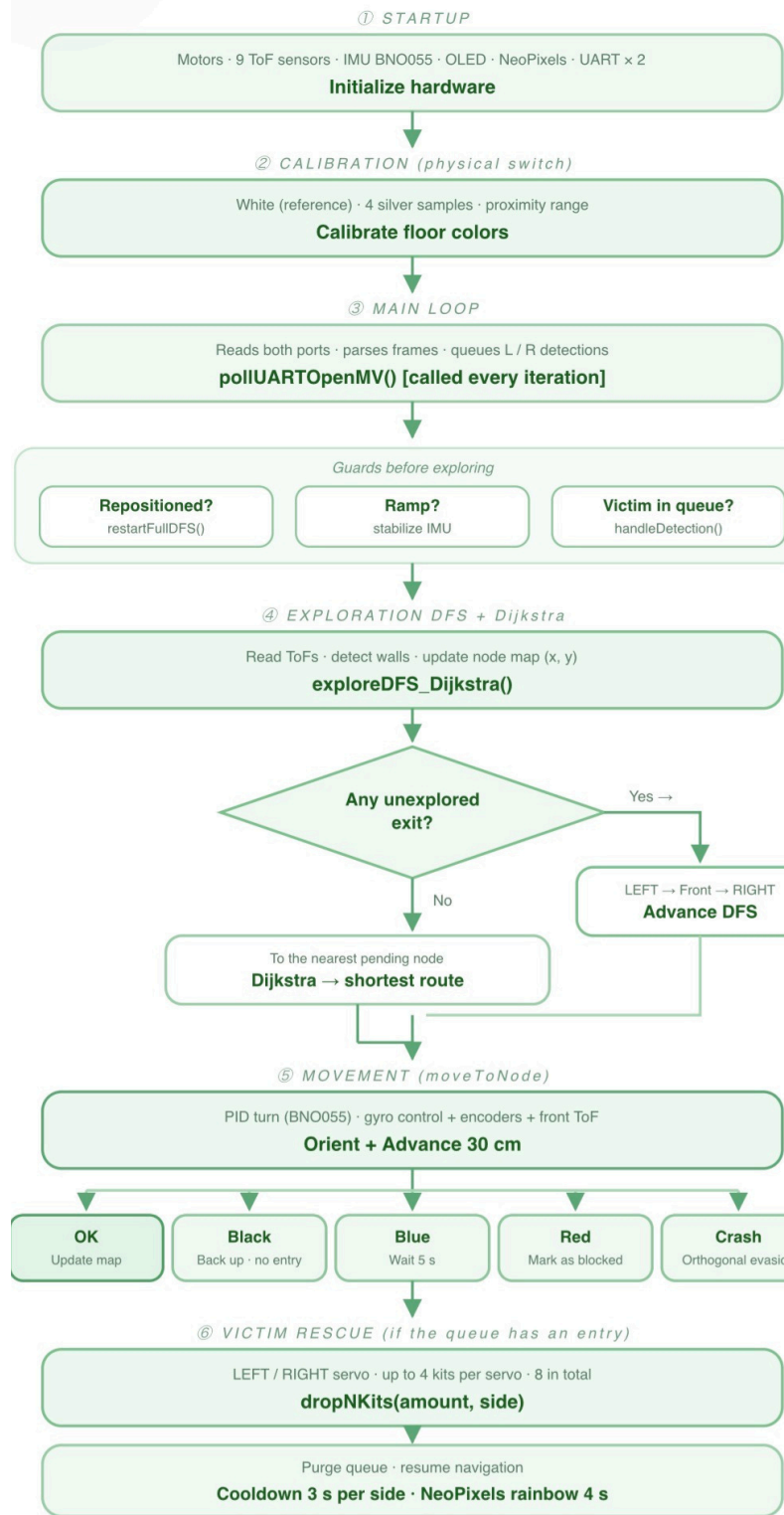
If a lack of progress occurs after a checkpoint, this objective is discarded, and the robot prioritizes exploring the entire map, including the red zone. The strategy aims to minimize the Lack of Progress (LoP) and maximize the score through adaptive decisions during execution.

Victim Detection and Delivery

An OpenMV camera handles real-time victim detection using a resource-efficient, dual-approach system:
Letter Victims: A lightweight MobileNetV2 neural network classifies alphabetical characters on the maze walls.
Cognitive Targets: A geometry-based Computer Vision algorithm processes targets without heavy neural networks through three strict stages:
Detection & Proximity Validation: Uses color blob detection and geometric filtering to identify perspective-distorted ellipses. To comply with the 15 cm deployment rule, analysis only triggers if the ellipse's minor axis exceeds a safe pixel threshold.
LAB Space Classification: Samples pixels across 5 concentric rings. By calculating the squared Euclidean distance to pre-calibrated LAB centroids, it classifies colors robustly against varying venue lighting: Blue (2), Green (1), Yellow (0), Red (-1), or Black (-2).
Health Status Evaluation: Sums the 5 ring values to determine the exact kit requirement (Harmed = 2, Stable = 1, Unharmed = 0). Invalid mathematical sums are instantly rejected as false positives.
2. Communication & Actuation
UART Protocol: The OpenMV encodes the required kit count (0-2) and transmits it to the main controller via UART (pins 2 & 5) for fast, reliable synchronization.
Inventory & Kit Management: The robot features two side towers holding 4 rescue kits each. Based on the UART signal and the target's location, the main controller activates Neopixel indicators, actuates the corresponding tower's servomotor to drop the kits, and updates its internal inventory counter accordingly.

Movement Functions

Advancement between tiles is achieved through controlled 30 cm movements, utilizing encoders to measure the distance traveled and ensure precision in every movement. To maintain the correct orientation, an IMU-based correction system is employed, allowing for precise 90° and 180° turns. Furthermore, position correction routines are implemented using distance and contact sensors, allowing the robot to align itself with the walls and reduce accumulated errors.



HARDWARE

Electronics

Teensy 4.1	Microcontroller responsible for receiving information from the sensors and executing movements.	APDS9960	RGB color sensor to detect black, blue, red, and silver floors.	TB6612FNG	H-bridge module to control the speed and direction of the DC motors.
OpenMV H7+	Programmable camera for computer vision. Responsible for victim detection.	BNO055	IMU that provides absolute orientation. Responsible for controlling turns and inclinations.	Micromotor N20	DC motors that offer good torque and speed. They feature encoders to calculate the distance traveled.
D24V50F5	High-current step-down voltage converter. Responsible for powering 5V components.	VL53L0X	Distance sensor with a range of up to 4m. Responsible for sensing walls.	Micro Limit Switch	Binary sensor that, along with the bumper design, allows for the detection of obstacles.
AMS1117	Low-dropout (LDO) linear voltage regulator. Responsible for powering 3.3V components.	WS2812	Programmable RGB LED strip/ring. Responsible for providing light and indicators during the round.	SG90	Limited-rotation servo motor. Along with a rack and pinion mechanism, it is responsible for dispensing kits.

PCB

The robot's electronic system is integrated through a custom PCB, designed to centralize connections and optimize space usage within the chassis. The board was developed considering the efficient distribution of all components, including sensors, actuators, and communication modules. The design also considers the proper management of communication protocols such as I2C and UART. This allows for reduced wiring and improved internal organization. Thanks to this integration, the PCB acts as the core of the robot, ensuring a robust and organized connection that is adaptable to future improvements.

Mechanics

The mechanical design of the robot is aimed at ensuring stable and adaptable navigation in complex environments. A compact structure with approximate dimensions of 16x14x14 cm was developed, allowing for better control within the maze. Additionally, a low and centered center of gravity was prioritized, achieving greater stability and reducing the risk of tipping over when overcoming obstacles such as ramps and speed bumps.

Suspension System

The suspension is designed as a mechanical system that allows a certain degree of rotational movement in the wheels through axles and movable joints. When the robot traverses irregularities such as speed bumps, ramps, or steps, the suspension absorbs these variations, allowing the wheels to adjust to the terrain. This prevents the loss of contact with the surface and maintains the robot's stability.

Bumper

The robot features a front bumper system made of flexible material, integrated with limit switches. This mechanism allows it to detect collisions with walls or obstacles, facilitating position correction and reducing accumulated errors during navigation.

Wheels

The tires were designed to maximize traction on different surfaces. They are made of silicone and feature a pattern that allows them to adapt to obstacles such as ramps and stairs, improving the robot's performance on variable terrain.

Kit Ejection System

The kit delivery system is based on a servo-driven rack and pinion mechanism, designed to be compact and precise.

